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|----------------------------------------------------------|-------------------------------------------------------|
| Programme: <b>Diploma in Cloud Computing and BigData</b> |                                                       |
| Course Code : <b>6272B</b>                               | Course Title: <b>Introduction to Machine Learning</b> |
| Semester : 6                                             | Credits: <b>4</b>                                     |
| Course Category: <b>Engineering Science</b>              |                                                       |
| Periods per week: <b>4 (L:4 T:0 P:0)</b>                 | Periods per semester: <b>60</b>                       |

### Course Objectives:

- Understand Artificial Intelligence Concepts
- Understand Machine Learning algorithms and their applications

### Course Prerequisites:

| Topic                            | Course Code | Course name                     | Semester |
|----------------------------------|-------------|---------------------------------|----------|
| Calculus and coordinate geometry | 1002        | Mathematics I                   | 1        |
| Introduction to IT Systems Lab   | 1008        | Problem Solving and Programming | 2        |
| Linear Algebra                   | 2002        | Mathematics II                  | 2        |

### Course Outcomes:

On completion of the course, the student will be able to:

| CO <sub>n</sub> | Description                                      | Duration (Hours) | Cognitive Level |
|-----------------|--------------------------------------------------|------------------|-----------------|
| CO1             | Outline the concepts of Machine Learning Systems | 12               | Understanding   |
| CO2             | Explain the Regression Techniques                | 12               | Understanding   |
| CO3             | Describe the Classification techniques           | 20               | Understanding   |
| CO4             | Summarize Artificial Neural Network              | 14               | Understanding   |

|  |             |   |  |
|--|-------------|---|--|
|  | Series Test | 2 |  |
|--|-------------|---|--|

### CO – PO Mapping

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1             | 3   | 2   |     |     |     |     |     |
| CO2             | 3   | 2   |     |     |     |     |     |
| CO3             | 3   | 2   |     |     |     |     |     |
| CO4             | 3   | 2   |     | 2   |     |     |     |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

### Course Outline

| Module Outcomes                                                                                                                                                                                                                                                                                                                                                      | Description                                                             | Duration (Hours) | Cognitive Level |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|------------------|-----------------|
| <b>CO1</b>                                                                                                                                                                                                                                                                                                                                                           | <b>Outline the concepts of Machine Learning Systems</b>                 |                  |                 |
| M1.01                                                                                                                                                                                                                                                                                                                                                                | Illustrate Supervised and Semi-supervised Learning with examples        | 2                | Understanding   |
| M1.02                                                                                                                                                                                                                                                                                                                                                                | Describe Unsupervised and Reinforcement Learning with examples          | 2                | Understanding   |
| M1.03                                                                                                                                                                                                                                                                                                                                                                | Summarize the applications of Classification, Prediction and Clustering | 4                | Understanding   |
| M1.04                                                                                                                                                                                                                                                                                                                                                                | Explain about Data Preparation                                          | 4                | Understanding   |
| <b>Contents:</b><br><b>Machine Learning Types-</b> Supervised, Semi-supervised, Unsupervised and Reinforcement Learning<br><b>Machine Learning Tasks-</b> Prediction, Classification and Clustering- Applications examples of each<br><b>Data Preparation-</b> Data Cleaning, Feature Scaling, Linear discriminant analysis (LDA), Principal Component Analysis(PCA) |                                                                         |                  |                 |
| <b>CO2</b>                                                                                                                                                                                                                                                                                                                                                           | <b>Explain the Regression Techniques</b>                                |                  |                 |
| M2.01                                                                                                                                                                                                                                                                                                                                                                | Describe the basic concepts in Linear Regression Model Prediction       | 3                | Understanding   |

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| M2.02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Explain about Gradient Descent, Batch Gradient Descent, Stochastic Gradient Descent and Mini-Batch Gradient Descent | 5 | Understanding |
| M2.03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Classify Regularized Linear Models- Ridge and Lasso , Early Stopping                                                | 2 | Understanding |
| M2.04                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Summarize the Evaluation Metrics                                                                                    | 2 | Understanding |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Series Test – I                                                                                                     | 1 |               |
| <b>Contents:</b> Linear Regression for Prediction- Applications, Model Parameters, Hyperparameters, Cost Function – MSE, Residuals, Gradient Descent -batch, mini-batch and stochastic<br>Simple Linear Regression application for prediction<br>Evaluation Metrics- Accuracy, Precision, Recall, Specificity, AUC, F-Measure                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                     |   |               |
| <b>CO3</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Describe the Classification Techniques</b>                                                                       |   |               |
| M3.01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Explain Logistic Regression                                                                                         | 6 | Understanding |
| M3.02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Illustrate the Decision Tree                                                                                        | 4 | Understanding |
| M3.03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Describe Random Forest                                                                                              | 4 | Understanding |
| M3.04                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Summarize about Support Vector Machine                                                                              | 6 | Understanding |
| <b>Contents:</b><br><b>Classification-</b> Binary and Multiclass, Logistic Regression- Training and Cost function, Cross Entropy, Advantages and disadvantages of Logistic Regression.<br><b>Decision Tree-</b> Training and Visualization of Decision Tree, Splitting Criteria, Information gain, Gini Index, Stopping Criteria, Pruning methods, Advantages and disadvantages of Decision tree.<br>RDF- Ensemble methods, majority voting, Hyperparameters of RDF, advantages and disadvantages of RDF<br><b>Support Vector Machine (SVM)</b> – Concept of support vectors, hyperplane, Linear and Nonlinear SVM Classification, Outliers, Polynomial- RBF kernel, Similarity Features, Advantages and disadvantages of SVM classifier. |                                                                                                                     |   |               |
| <b>CO4</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Summarize Artificial Neural Network                                                                                 |   |               |
| M4.01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Explain Artificial Neural Network                                                                                   | 4 | Understanding |
| M4.02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Describe the concepts in Convolutional Neural Network                                                               | 6 | Understanding |

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| M4.03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Outline Recurring Neural Network | 2 | Understanding |
| M4.04                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Describe about Transfer Learning | 2 | Understanding |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Series Test – II                 | 1 |               |
| <p><b>Contents :</b> Biological Neurons, Artificial Neurons, Logical computations with neurons, The Perceptron, Multilayer perceptron(MLP), Backpropagation, Regression and classification MLPs</p> <p>Deep Learning Concepts, Convolutional Neural Network (CNN), Filters- Convolution and Pooling operation, Activation function, Applications of CNN</p> <p>Recurring Neural Network (RNN) concepts, memory cells, Applications of RNN- Natural language Processing</p> <p>Transfer Learning- CNN architectures- AlexNet, GoogleNet, VGG, ResNet and Xception</p> |                                  |   |               |